

Luke S. Snyder

✉ snyderl@cs.washington.edu <https://luke-s-snyder.github.io/>

EDUCATION

2021-2027 (expected)	Ph.D. , Computer Science & Engineering Advisor: Jeffrey Heer	University of Washington
2021-2023	M.S. , Computer Science & Engineering <i>DIVI: Dynamically Interactive Visualization</i> Advisor: Jeffrey Heer, Second Reader: Leilani Battle	University of Washington
2018-2020	M.S. , Computer Science <i>Predictive Visual Analytics of Social Media Data for Supporting Real-time Situational Awareness</i> Advisor: David Ebert Committee: Dan Goldwasser, Jeremiah Blocki	Purdue University
2014-2018	B.S. , Computer Science (Honors), Minor in Mathematics, <i>Summa Cum Laude</i>	University of Arkansas

PUBLICATIONS

Journal Papers (peer-reviewed)

- J6. **L. S. Snyder**, M. C. Stone, and J. Heer. Crossing the chasm : Bridging visual augmentations and designer intent. *IEEE Transactions on Visualization and Computer Graphics*, 32(7):5608–5625, July 2026
- J5. **L. S. Snyder** and J. Heer. Divi : Dynamically interactive visualization. *IEEE Transactions on Visualization and Computer Graphics*, 30(1):403–413, Jan 2024
- J4. M. M. Chiu, A. Morakhovski, D. Ebert, A. Reinert, and **L. S. Snyder**. Detecting covid-19 fake news on twitter : Followers, emotions, relationships, and uncertainty. *American Behavioral Scientist*, May 2023
- J3. A. Reinert, **L. S. Snyder**, J. Zhao, A. S. Fox, D. F. Hougen, C. Nicholson, and D. S. Ebert. Visual analytics for decision-making during pandemics. *Computing in Science & Engineering*, 22(6):48–59, Nov 2020
- J2. **L. S. Snyder**, Y.-S. Lin, M. Karimzadeh, D. Goldwasser, and D. S. Ebert. Interactive learning for identifying relevant tweets to support real-time situational awareness. *IEEE Transactions on Visualization and Computer Graphics*, 26(1):558–568, Jan 2020
- J1. J. Zhao, M. Karimzadeh, **L. S. Snyder**, C. Surakitbanharn, Z. C. Qian, and D. S. Ebert. MetricsVis: A visual analytics system for evaluating employee performance in public safety agencies. *IEEE Transactions on Visualization and Computer Graphics*, 26(1):1193–1203, Jan 2020

Conference Papers (peer-reviewed)

- C5. **L. S. Snyder**, C. Wang, and S. M. Drucker. Challenges & opportunities with llm-assisted visualization retargeting. *IEEE Visualization Short Papers (VIS)*, 2025
- C4. **L. S. Snyder**, R. A. Rossi, E. Koh, J. Heer, and J. Hoffswell. Interaction techniques for exploratory data visualization on mobile devices. *EuroVis 2024 - Short Papers*. The Eurographics Association, 2024
- C3. **L. S. Snyder**, J. Zhao, A. Reinert, G. Wang, and D. S. Ebert. PanViz 2.0: AI-driven visual analytics to adapt to the novel challenges of covid-19. *Proceedings of the Hawaii International Conference on System Sciences, HICSS-54*, Jan 2021
- C2. **L. S. Snyder**, M. Karimzadeh, C. Stober, and D. S. Ebert. Situational awareness enhanced through social media analytics: A survey of first responders. *IEEE International Symposium on Technologies for Homeland Security, HST 2019*, pp. 1–8. IEEE, Nov 2019
- C1. **L. S. Snyder**, M. Karimzadeh, R. Chen, and D. S. Ebert. City-level geolocation of tweets for real-time visual analytics. *Proceedings of the 3rd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery, GeoAI 2019*, pp. 85–88. ACM, Nov 2019

Book Chapters

- B1. M. Karimzadeh, J. Zhao, G. Wang, **L. S. Snyder**, and D. S. Ebert. Human-guided visual analytics for big data. *Big Data in Psychological Research*, pp. 145–177. American Psychological Association, Washington, DC, USA, Jan 2020

Research and Technical White Papers

T1. M. Karimzadeh, **L. S. Snyder**, and D. S. Ebert. Geovisual analytics and interactive machine learning for situational awareness. *US National Report to the International Cartographic Association*, ICC 2019. Tokyo, Japan, 2019

RESEARCH EXPERIENCE

University of Washington IDL Lab

Graduate Research Assistant with Jeffrey Heer

2021-Present

2025-Present **Augmenting scientific literature sensemaking**

Computational workflow & algorithms to distill accretive ideas across scholarly literature.

2023-2025 **Crossing the chasm : Briding visual augmentations and designer intent**

Design space of visual augmentations to bridge higher-level communication aims in data visualization.

Relevant publications : J6

2021-2023 **Divi : Dynamically interactive visualization**

Automatic orchestration of interactions within static SVG charts via runtime deconstruction.

Relevant publications : J5

Microsoft Research

Research Scientist Intern with Steven Drucker & Chenglong Wang

2024

2024 **Challenges & opportunities with llm-assisted visualization retargeting**

Pipeline to retarget published charts to new datasets via static program analysis and LLM code generation.

Relevant publications : C5

Adobe Research

Research Scientist Intern with Jane Hoffswell & Ryan Rossi

2022

2022 **Interaction techniques for exploratory data visualization on mobile devices**

Design probe for new exploratory interactions to augment data visualization on mobile devices.

Relevant publications : C4

University of Oklahoma DISC Institute

Research Associate with David Ebert

2020-2021

2020-2021 **Visual analytics for decision-making during pandemics**

Human-in-the-loop techniques for decision-making during COVID-19, including interactive interdiction planning and human-guided parameter inference for disease spread models.

Relevant publications : J4, J3, C3

Purdue University VACCINE Lab

Graduate Research Assistant with David Ebert

2018-2020

2018-2020 **Interactive learning for identifying relevant tweets to support real-time situational awareness**

Interactive machine learning framework and system to rapidly identify relevant information from streaming text data to facilitate real-time situational awareness.

Relevant publications : J2, T1, P1

2019 **City-level geolocation of tweets for real-time visual analytics**

Visual analytics system for real-time geolocation inference of non-geotagged streaming text data to support situational awareness during large-scale events.

Relevant publications : C1, P2

2019 **Situational awareness enhanced through social media analytics : A survey of first responders**

Survey on first responders' use of social media data, barriers and gaps between existing social media analytic systems and current research, and opportunities for future research directions.

Relevant publications : C2

TEACHING AND MENTORING

Spring 2023-2026	TA for Data Visualization at University of Washington
2023-2024	Graduate Mentor for Andrew Zhang at University of Washington (BS/MS Thesis)
2018-2020	Graduate Mentor for four undergraduate students participating in the Vertically Integrated Projects (VIP) at Purdue University
Summer 2019	Graduate Mentor for two students in the Summer Undergraduate Research Fellowship (SURF) program at Purdue University
2015-2016	Tutor for University Physics I, University Chemistry I and II, College Algebra, Plane Trigonometry, and Calculus I and II at University of Arkansas Class+ Center

SELECTED HONORS AND AWARDS

2020	Nominated by Purdue University for Midwestern Association of Graduate Schools Distinguished Masters Thesis Award
2018	First-ranked Senior Scholar awarded by University of Arkansas
2018	Most Outstanding Senior in Computer Science awarded by University of Arkansas
2018	Honors College Research Grant awarded by University of Arkansas
2018	J.B. Hunt Hackathon Winner
2014-2018	Chancellor and Dean's List awarded by University of Arkansas
2015	Freshman of the Year Finalist awarded by the University of Arkansas College of Engineering

PROFESSIONAL SERVICES

Organizations

Secretary, Association for Computing Machinery (ACM), University of Arkansas, 2016-2017
Honors College Ambassador, University of Arkansas, 2015-2018

Conference Reviewer

User Interface Software and Technology (UIST 2026)
The IEEE Conference on Visual Analytics Science and Technology (IEEE VAST 2019)
Hawaii International Conference on System Sciences (HICSS-53, 54)